

Stochastic Models of Higher-Order Networks

Point Processes and Topological Data Analysis

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Copenhagen, Denmark

15 December 2025

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- 1 Poisson Approximation in Weighted Random Connection Models
- 2 Random Connection Hypergraphs
- 3 Functional Convergence of Edge Counts in Dynamic RCMs

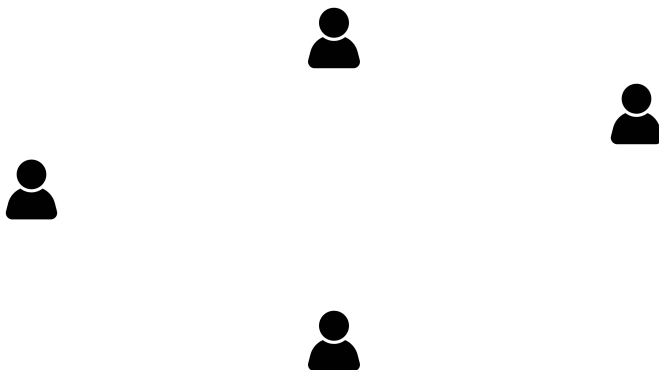
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- (Topology of Higher-Order Random Connection Models)

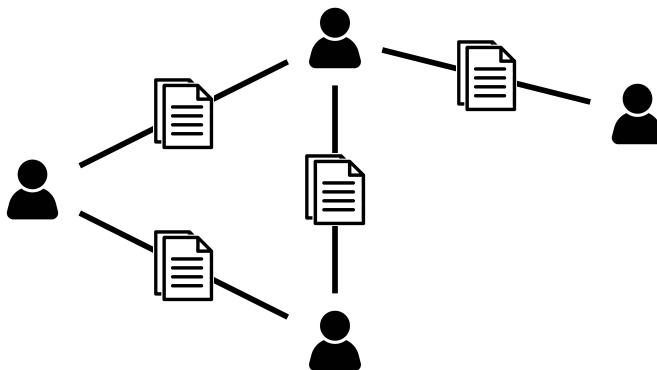
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Complex Systems



Complex Systems

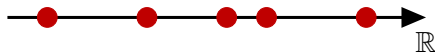


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 - power-law distribution
 - small-world property
 - community structure (clustering)

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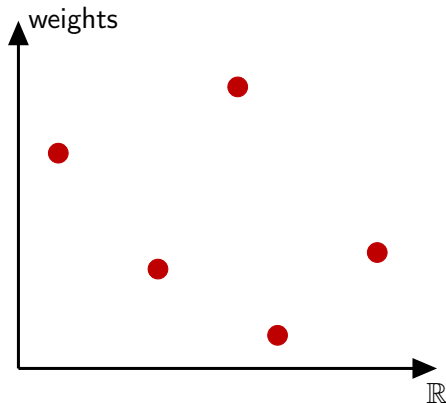
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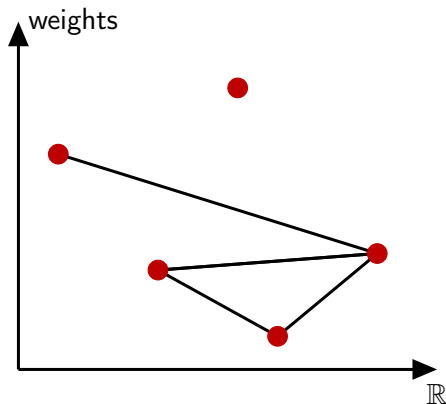
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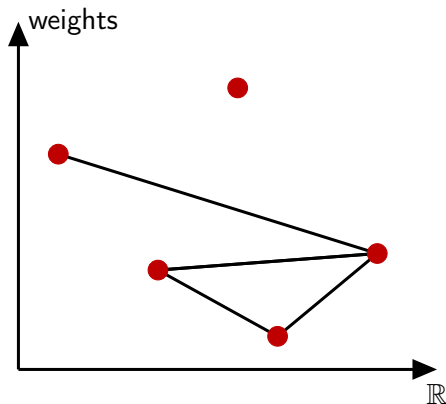
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M. Deijfen, R. van der Hofstad, and G. Hooghiemstra. Scale-free percolation. *Ann. Inst. Henri Poincaré Probab. Stat.*, 49(3):817–838, 2013.

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- connection probability: $\varphi: [0, \infty) \rightarrow [0, 1]$

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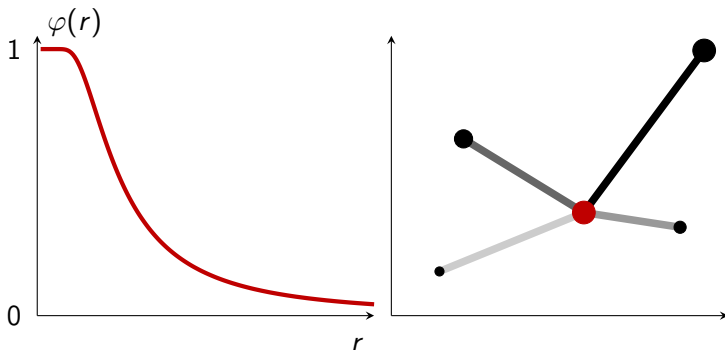
$$\varphi\left(\frac{|B_{|x-y|}(o)|}{v_s \kappa(W_x, W_y)}\right) \int_0^\infty \varphi(r) dr = 1$$

- regularity conditions on φ and F_W :

$$\lim_{r \uparrow \infty} \frac{\varphi(tr)}{\varphi(r)} = t^{-\alpha} \quad \lim_{w \uparrow \infty} \frac{1 - F_W(tw)}{1 - F_W(w)} = t^{-\beta} \quad \alpha > 1, \beta \geq a\alpha$$

Example

$$\varphi(r) := 1 - \exp(-r^{-2}/\pi) \quad \kappa(w_1, w_2) := w_1 w_2$$



Goal of the Study

- Spatial distribution of k -degree vertices:

$$\xi_s := \sum_{x \in \mathcal{P}_s \cap [0,1]^d} \mathbb{1}\{\deg(x) = k\} \delta_x$$

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- Specify scaling v_s such that $\mathbb{E}[\xi_s([0,1]^d)] = 1$
- Typical weight of a degree- k node:

$$\mathbb{P}(W \leq w_s) = 1/(2s)$$

Poisson Approximation

Theorem (Hirsch, Jahnelt, Jhavar, J.; 2025)

ζ : Poisson point process with intensity $\mathbb{1}\{x \in [0, 1]^d\}$

If

$$\begin{aligned} (sv_s)^{-1} w_s^{-\eta} &\in o(1/\log(s)) \\ F(w_s^\eta) w_s^{-(K+1)(1-\eta)} &\in o(1/\log(s)) \\ w_s^{-(1-K\delta)(1-\eta)} &\in o(1/\log(s)). \end{aligned}$$

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Then $d_{KR}(\xi_s, \zeta) = \sup_{h \in \text{LIP}(1)} \left\{ |\mathbb{E}[h(\xi_s)] - \mathbb{E}[h(\zeta)]| \right\} \xrightarrow{s \uparrow \infty} 0$

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M. D. Penrose. Inhomogeneous random graphs, isolated vertices, and Poisson approximation. *J. Appl. Probab.*, 55(1):112–136, 2018.

S. K. Iyer and S. K. Jhavar. Poisson approximation and connectivity in a scale-free random connection model. *Electron. J. Probab.*, 26(23):Paper No. 86, 2021.

Stein's Method

O. Bobrowski, M. Schulte, and D. Yogeshwaran. Poisson process approximation under stabilization and Palm coupling. *Ann. H. Lebesgue*, 5:1489–1534, 2022.

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 $\{\omega \in \mathbf{N}_{\mathbb{X}}: \mathcal{S}(\mathbf{x}, \omega) \subseteq A\} = \{\omega \in \mathbf{N}_{\mathbb{X}}: \mathcal{S}(\mathbf{x}, \omega \cap A) \subseteq A\} \quad (\forall A \subseteq \mathbb{X})$

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- Function
 $f: \mathbb{X}^m \times \mathbf{N}_{\mathbb{X}} \rightarrow \mathbb{Y} \quad f(\mathbf{x}, \omega) = f(\mathbf{x}, \omega \cap A) \quad \forall A \supset \mathcal{S}(\mathbf{x}, \omega)$

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Theorem (Bobrowski, Schulte, Yogeshwaran; 2022)

- η, ζ : *Poisson point processes on $\mathbb{X}$ with intensities $\lambda_\eta, \lambda_\zeta < \infty$*

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Then

$$d_{\text{KR}}(\xi, \zeta) \leq d_{\text{TV}}(\lambda_\xi, \lambda_\zeta) + E_1 + E_2 + E_3 + E_4$$

Error Terms

Theorem (Bobrowski, Schulte, Yogeshwaran; 2022)

$$\hat{\mathcal{S}} : \mathbb{X}^m \rightarrow \mathcal{F} \quad \mathbf{x} \subseteq \hat{\mathcal{S}}(\mathbf{x}) \quad g_{\text{tr}}(\mathbf{x}, \omega) := g(\mathbf{x}, \omega) \mathbb{1}\{\mathcal{S}(\mathbf{x}, \omega) \subseteq \hat{\mathcal{S}}(\mathbf{x})\}$$

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$$E_1 := \frac{2}{m!} \int_{\mathbb{X}^m} \mathbb{E}[g(\mathbf{x}, \eta \cup \mathbf{x}) \mathbb{1}\{\mathcal{S}(\mathbf{x}, \eta \cup \mathbf{x}) \not\subseteq \hat{S}(\mathbf{x})\}] \lambda_{\xi}^{\otimes m}(\mathrm{d}\mathbf{x})$$

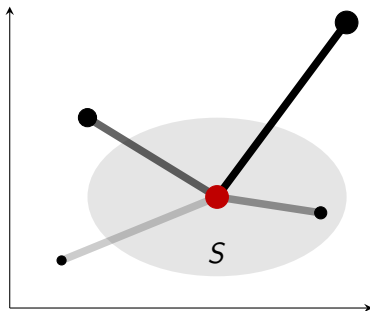
$$E_2 := \frac{2}{(m!)^2} \iint_{\mathbb{X}^m \times \mathbb{X}^m} \mathbb{1}\{\hat{S}(\mathbf{x}) \cap \hat{S}(\mathbf{z}) \neq \emptyset\} \\ \times \mathbb{E}[g_{\text{tr}}(\mathbf{x}, \eta \cup \mathbf{x})] \mathbb{E}[g_{\text{tr}}(\mathbf{z}, \eta \cup \mathbf{z})] \lambda_{\xi}^{\otimes m}(\mathrm{d}\mathbf{x}) \lambda_{\xi}^{\otimes m}(\mathrm{d}\mathbf{z})$$

$$E_3 := \frac{2}{(m!)^2} \iint_{\mathbb{X}^m \times \mathbb{X}^m} \mathbb{1}\{\hat{S}(\mathbf{x}) \cap \hat{S}(\mathbf{z}) \neq \emptyset\} \\ \times \mathbb{E}[g_{\text{tr}}(\mathbf{x}, \eta \cup \mathbf{x} \cup \mathbf{z}) g_{\text{tr}}(\mathbf{z}, \eta \cup \mathbf{x} \cup \mathbf{z})] \lambda_{\xi}^{\otimes m}(\mathrm{d}\mathbf{x}) \lambda_{\xi}^{\otimes m}(\mathrm{d}\mathbf{z})$$

$$E_4 := \frac{2}{m!} \sum_{\emptyset \subsetneq I \subsetneq \{1, \dots, m\}} \frac{1}{(m - |I|)!} \int_{\mathbb{X}^m} \int_{\mathbb{X}^{m - |I|}} \mathbb{E}[g_{\text{tr}}(\mathbf{x}, \eta \cup \mathbf{x} \cup \mathbf{z}) \\ \times g_{\text{tr}}((\mathbf{x}_I, \mathbf{z}), \eta \cup \mathbf{x} \cup \mathbf{z})] \lambda_{\xi}^{\otimes m - |I|}(\mathrm{d}\mathbf{z}) \lambda_{\xi}^{\otimes m}(\mathrm{d}\mathbf{x})$$

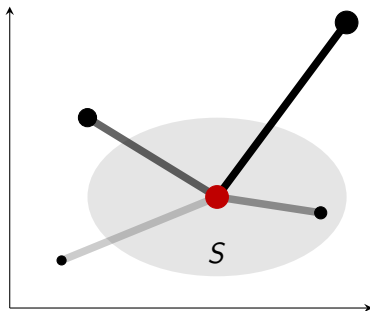
Proof Sketch

- Problem: g should be localized to $S \in \mathcal{F}$



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- Step 1: mark approximation
neglect high weighted points
- Step 2: reach approximation:
neglect far points

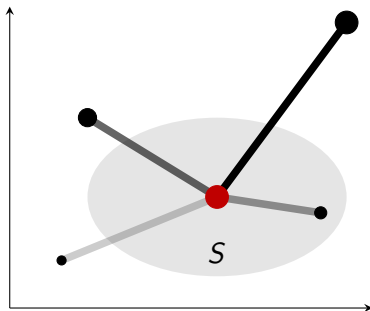
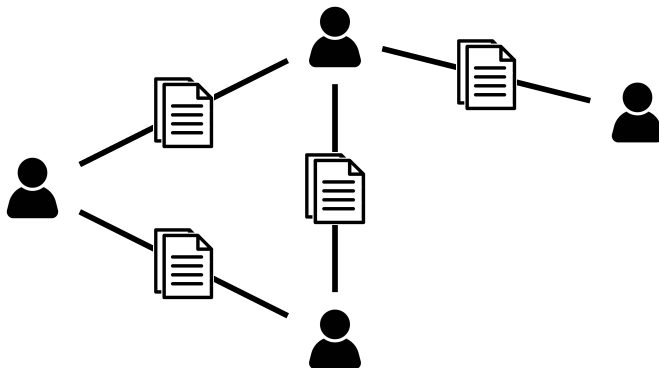


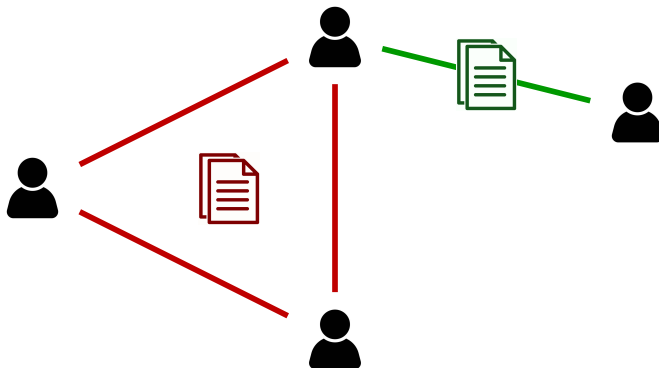
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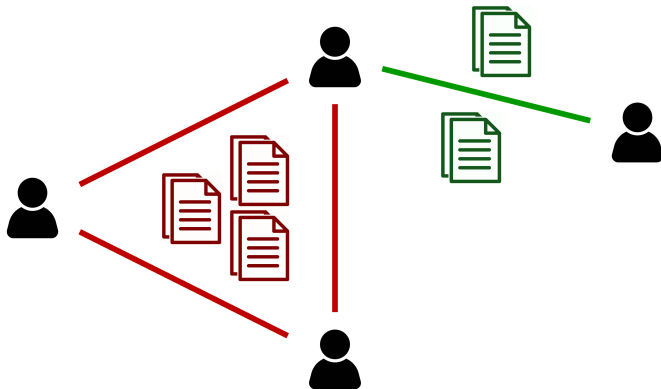
Complex Systems



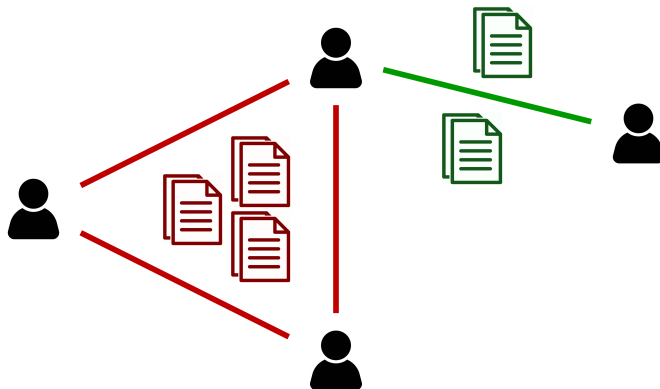
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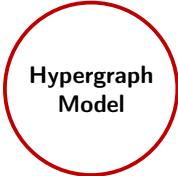


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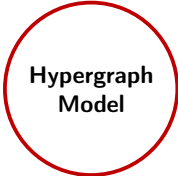
Simple graphs: information loss

Outline



Hypergraph Model

Outline

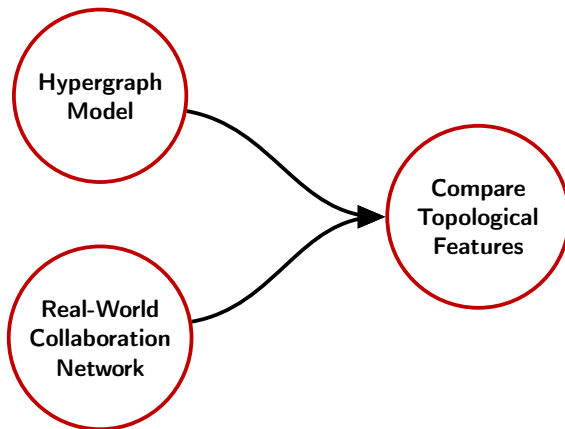


**Hypergraph
Model**



**Real-World
Collaboration
Network**

Outline



Random Connection Hypergraph Model (RCHM)

Two Poisson point processes

on $\mathbb{S} := \mathbb{R} \times [0, 1]$

- vertices: $\mathcal{P} \subset \mathbb{S}$
- hyperedges: $\mathcal{P}' \subset \mathbb{S}$

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Connections:

$$\{(X, U), (Y, V)\} \in \mathcal{P} \times \mathcal{P}'$$

$$|X - Y| \leq \beta U^{-\gamma} V^{-\gamma'} \quad \gamma, \gamma' \in (0, 1)$$

Random Connection Hypergraph Model (RCHM)

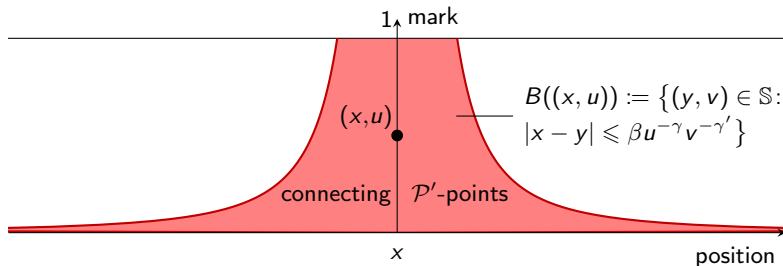
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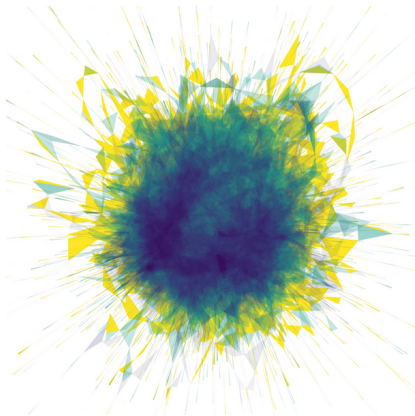
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R. van der Hofstad, J. Komjáthy, and V. Vadon. Random intersection graphs with communities. *Adv. in Appl. Probab.*, 53(4):1061–1089, 2021.

Collaboration Network

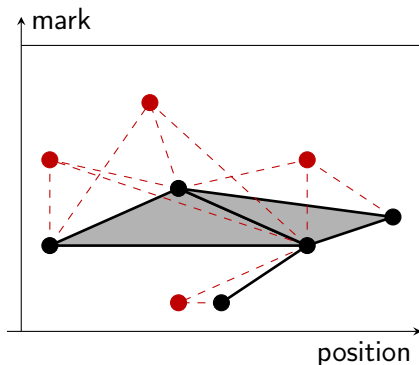
- Source: arXiv preprint server
- Vertices: $\sim 8 \cdot 10^5$ authors
- Hyperedges: $\sim 10^6$ documents



Statistics dataset

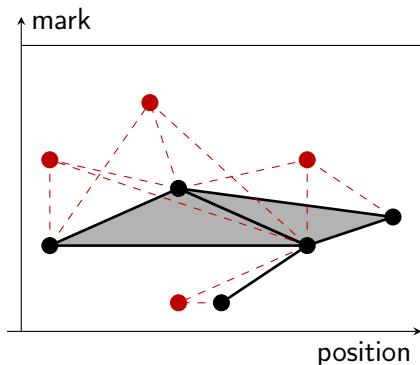
Higher-Order Degree Distribution

$$\deg(\Delta) := \mathcal{P}'\left(\bigcap_{p \in \Delta} B(p)\right)$$

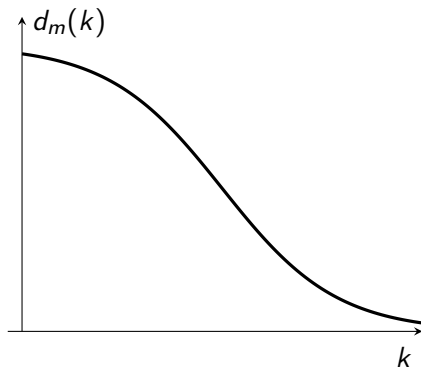


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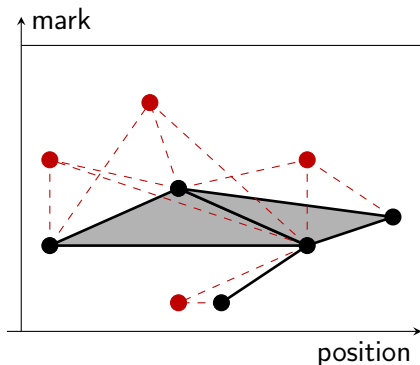


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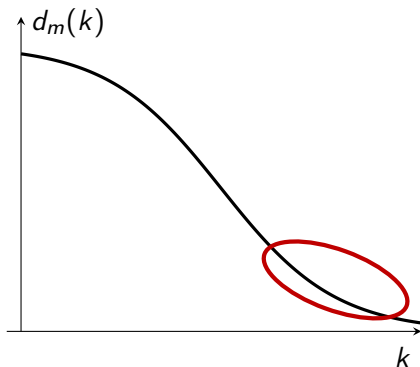


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Higher-Order Degree Distributions

Theorem (Brun, Hirsch, J., Otto; 2025+)

$$\lim_{k \uparrow \infty} \log(d_m(k)) / \log(k) = m - (m + 1)/\gamma,$$

where

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- Parameter fitting: model parameters can be tuned to obtain a given power-law exponent.

Simplicial Complex

Idea: represent higher-order relationships with a simplicial complex

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Definition (Simplicial Complex)

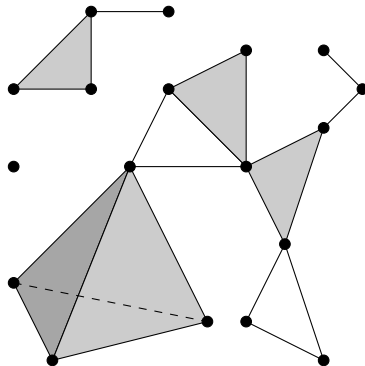
- (P, Σ)
- $P \neq \emptyset, |P| < \infty$
- $\Sigma \subseteq 2^P$
- $p \in P \implies \{p\} \in \Sigma$
- $\sigma \in \Sigma, \Delta \subseteq \sigma \implies \Delta \in \Sigma$

Simplicial Complex

Idea: represent higher-order relationships with a simplicial complex

Definition (Simplicial Complex)

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Simplicial Complex

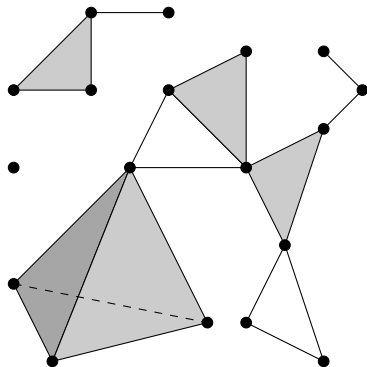
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RCHM = simplicial complex

$$\Sigma = \{\Delta \subseteq \mathcal{P} : \deg(\Delta) \geq 1\}$$



Betti Numbers

Definition (k -chains)

$$C_k(K) := \left\{ \sum_i a_i \sigma_i, : a_i \in \mathbb{Z}_2, \sigma_i \in \Sigma, |\sigma_i| = k + 1 \right\}$$

Betti Numbers

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Definition (Boundary operator)

$$\partial_k: C_k(K) \rightarrow C_{k-1}(K) \qquad \partial_k(\sigma_i) = \sum_{j=0}^k \sigma_i^{(j)}$$

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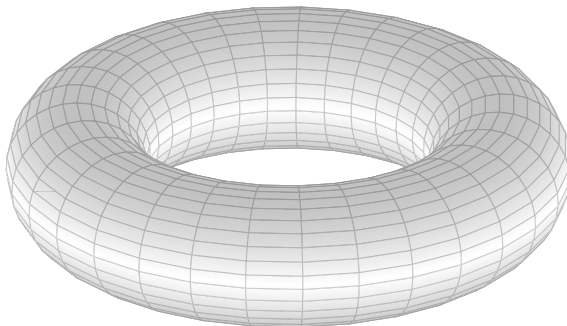
Definition (Cycle and boundary groups)

Cycle group: $Z_k = \ker(\partial_k)$

Boundary group: $B_k = \text{im}(\partial_{k+1})$

k th Betti number: $\beta_k = \dim(Z_k) - \dim(B_k)$

Betti Numbers



$$\beta_0 = 1; \beta_1 = 2; \beta_2 = 1$$

CLT for Betti Numbers

- Window: $\mathbb{S}_n := [0, n] \times [0, 1]$
- Simplicial complex: $G_n := G(\mathcal{P} \cap \mathbb{S}_n, \mathcal{P}' \cap \mathbb{S}_n)$
- Betti number of dimension m in window n : $\beta_m^{(n)} := \beta_m(G_n)$

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Theorem (Brun, Hirsch, J., Otto; 2025+)

If $\gamma < 1/4$, $\gamma' < 1/(4(m+2))$, then

$$n^{-1/2}(\beta_m^{(n)} - \mathbb{E}[\beta_m^{(n)}]) \xrightarrow[n \uparrow \infty]{d} \mathcal{N}(0, \sigma^2) \quad \sigma^2 \geq 0.$$

CLT for Betti Numbers

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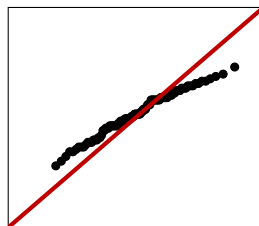
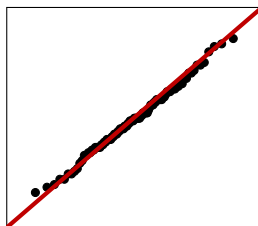
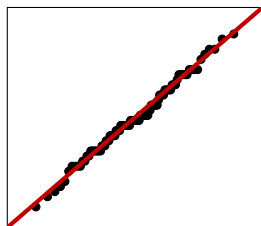
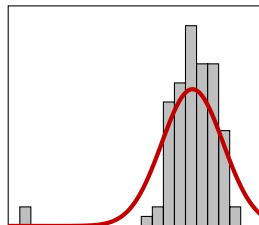
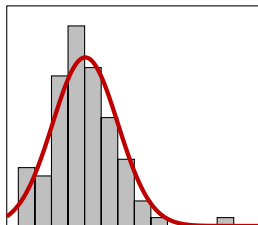
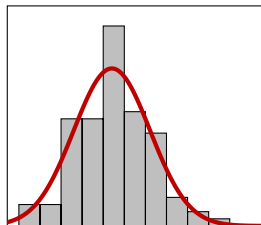
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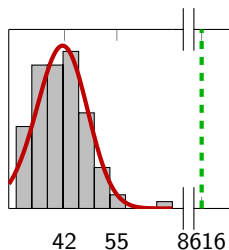
M. D. Penrose and J. E. Yukich. Central limit theorems for some graphs in computational geometry. *Ann. Appl. Probab.*, 11(4):1005–1041, 2001.

D. Pabst. Betti numbers in the random connection model for higher-dimensional simplicial complexes and the Boolean model. *arXiv preprint arXiv:2506.13429*, 2025.

Simulated Distribution of Betti-1 Numbers

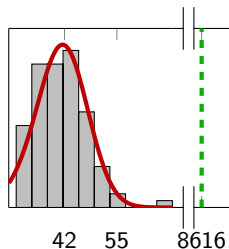
 $\gamma = 0.25$ $\gamma = 0.50$ $\gamma = 0.75$

Hypothesis Test

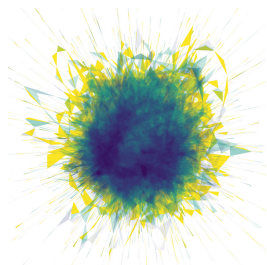


Hypothesis test

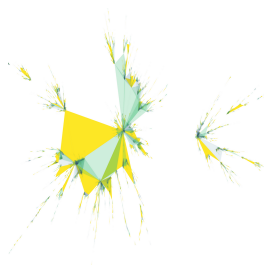
Hypothesis Test



Hypothesis test



Network data

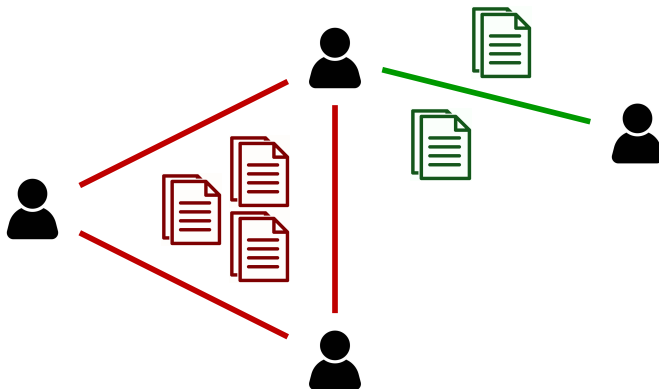


Simulation

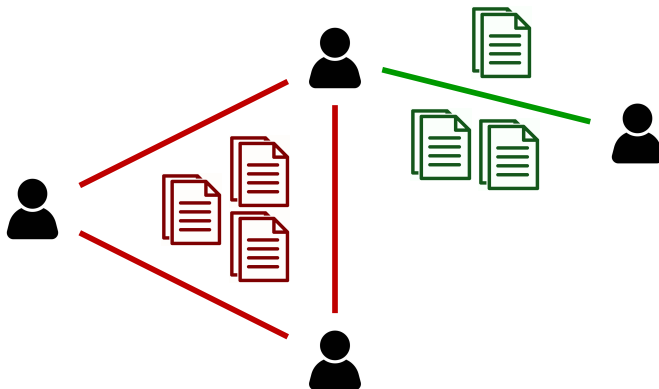
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- 1 Poisson Approximation in Weighted Random Connection Models
- 2 Random Connection Hypergraphs
- 3 Functional Convergence of Edge Counts in Dynamic RCMs

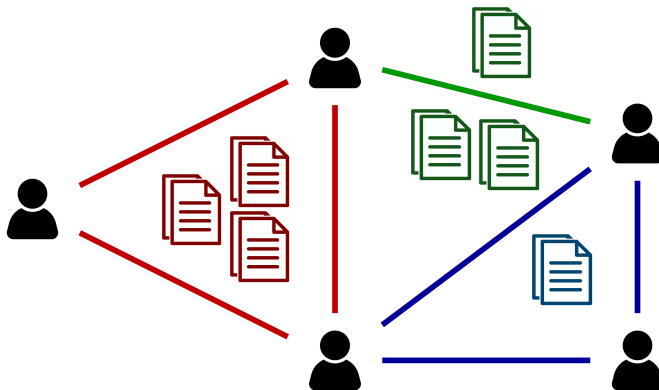
Complex Systems



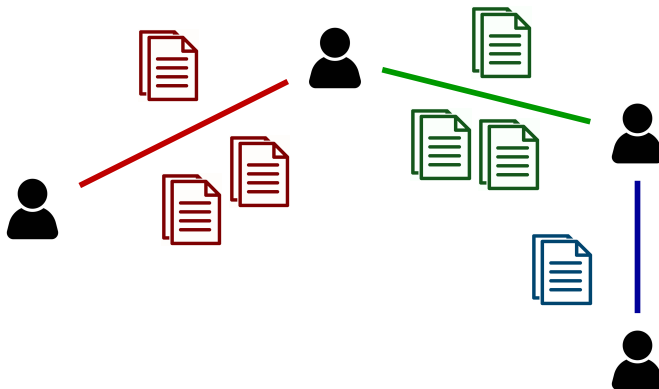
Complex Systems



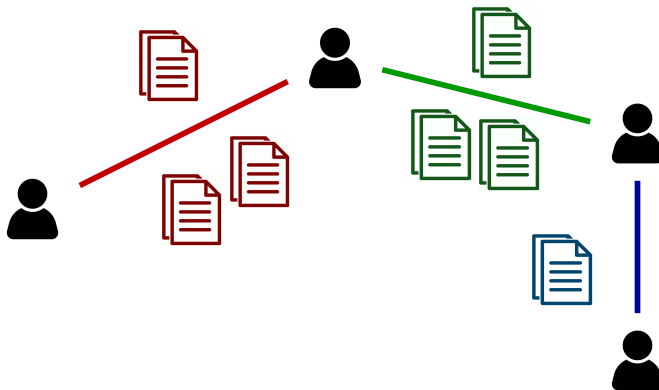
Complex Systems



Complex Systems



Complex Systems



Network dynamics?

Dynamic Random Connection Hypergraph Model

Vertices: $\mathcal{P}$

position	$X \in \mathbb{R}$	Leb
mark	$U \in (0, 1]$	Leb
birth time	$B \in \mathbb{R}$	Leb
lifetime	$L \in [0, \infty)$	Exp(1)

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Connection conditions:

spatial condition

$$|X - Z| \leq \beta U^{-\gamma} W^{-\gamma'}$$

temporal condition

$$B \leq R \leq B + L$$

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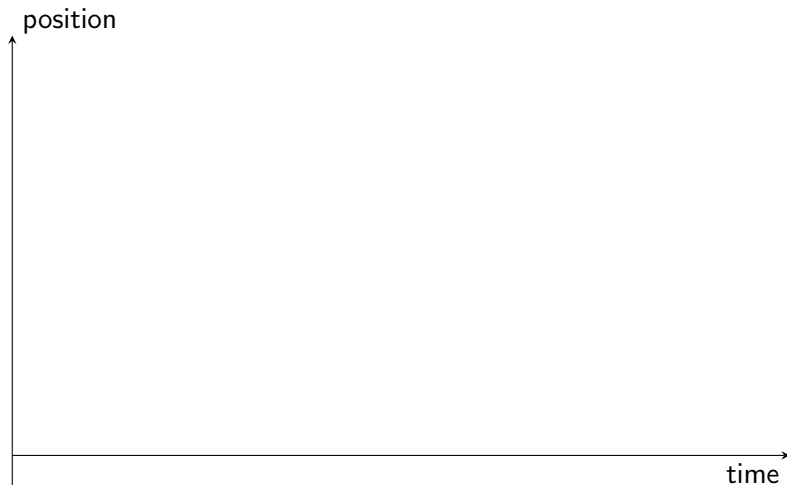
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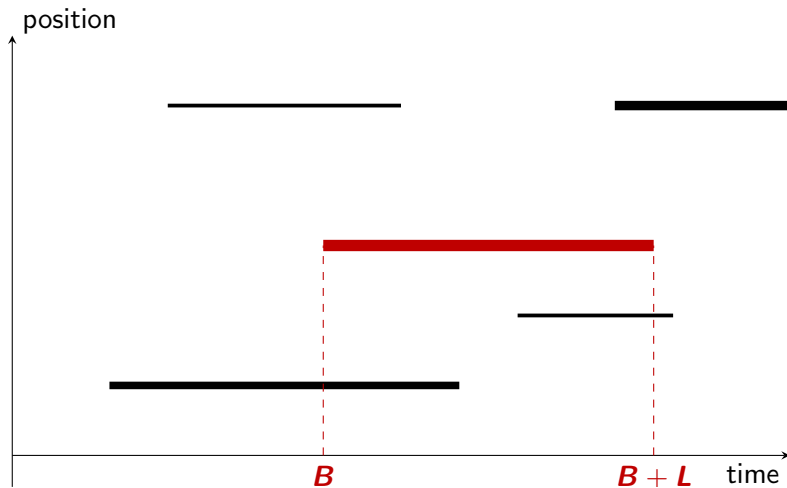
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E. Onaran, O. Bobrowski, and R. J. Adler. Functional central limit theorems for local statistics of spatial birth–death processes in the thermodynamic regime. *Ann. Appl. Probab.*, 33(5):3958–3986, 2023.

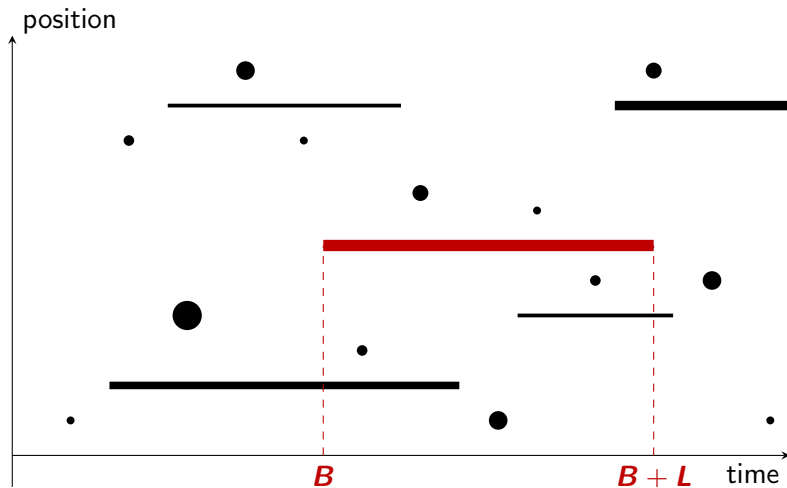
Dynamic Random Connection Hypergraph Model



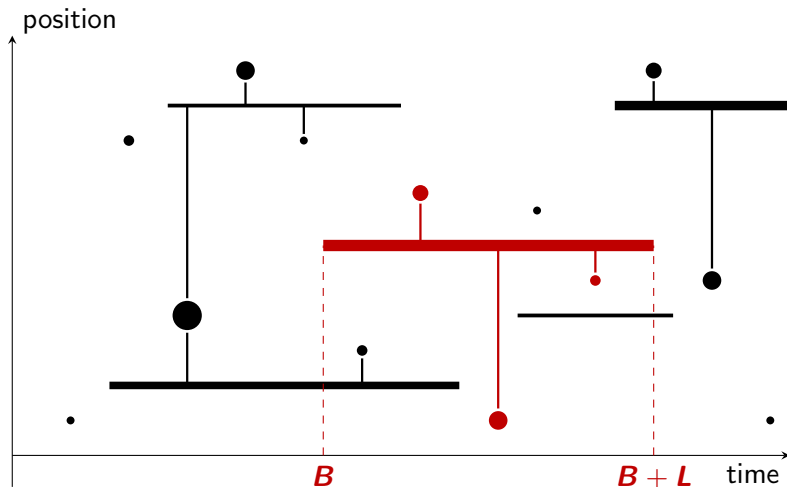
Dynamic Random Connection Hypergraph Model



Dynamic Random Connection Hypergraph Model



Dynamic Random Connection Hypergraph Model



Edge counts

- Degree of $P := (X, U, B, L) \in \mathcal{P}$:

$$\begin{aligned} \deg(P; t) := & \sum_{(Z, W, R) \in \mathcal{P}'} \mathbb{1}\{|X - Z| \leq \beta U^{-\gamma} W^{-\gamma'}\} \\ & \times \mathbb{1}\{B \leq R \leq t \leq B + L\} \end{aligned}$$

Edge counts

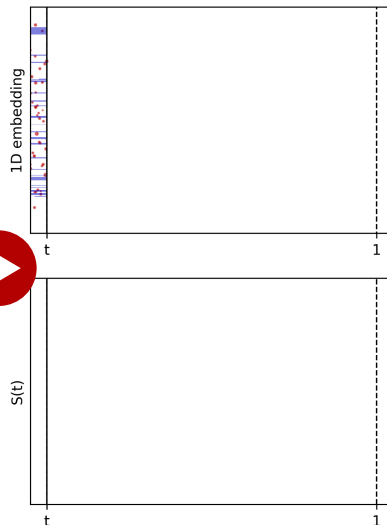
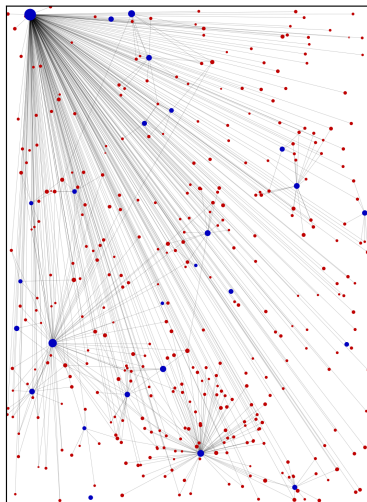
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- Edge count

$$S_n(\cdot) := \sum_{P \in \mathcal{P}} \deg(P; \cdot) \mathbb{1}\{X \in [0, n]\}$$

Edge counts



Finite Variance Case

Proposition (Hirsch, Jahnel, J.; 2025+)

If $\gamma < 1/2$ and $\gamma' \in (0, 1)$, then

$$\bar{S}_n(t) := n^{-1/2}(S_n(t) - \mathbb{E}[S_n(t)]) \xrightarrow[n \uparrow \infty]{d} \mathcal{N}(0, \sigma^2) \quad \sigma^2 > 0.$$

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Theorem (Hirsch, Jahnel, J.; 2025+)

G : Gaussian process

$$\text{Cov}(G(t_1), G(t_2)) = (c_1 + c_2|t_1 - t_2|) \exp(-\mu|t_1 - t_2|)$$

If $\gamma, \gamma' < 1/4$, then

$$\bar{S}_n(\cdot) \xrightarrow[n \uparrow \infty]{d} G(\cdot).$$

Univariate Stable Limit Theorem

Theorem (Hirsch, Jahnel, J.; 2025+)

If $\gamma > 1/2$, then

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Functional Convergence of Edge Counts

- $\nu([\varepsilon, \infty)) := c\varepsilon^{-1/\gamma}$: Lévy measure on $[0, \infty)$
- $\mathcal{P}_\infty := \text{PPP}(\nu \otimes \text{Leb}_{\mathbb{R}} \otimes \text{Exp}(1))$: Poisson point process on $[0, \infty) \times \mathbb{R} \times \mathbb{R}_+$

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$$S_\varepsilon^*(\cdot) := \sum_{(J,B,L) \in \mathcal{P}_\infty} J(\cdot - B) \mathbb{1}\{J \geq \tilde{c}\varepsilon^\gamma\} \mathbb{1}\{B \leq \cdot \leq B + L\}$$

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$$\bar{S}_\infty(\cdot) := \lim_{\varepsilon \downarrow 0} \bar{S}_\varepsilon^*(\cdot) := \lim_{\varepsilon \downarrow 0} (S_\varepsilon^*(\cdot) - \mathbb{E}[S_\varepsilon^*(\cdot)])$$

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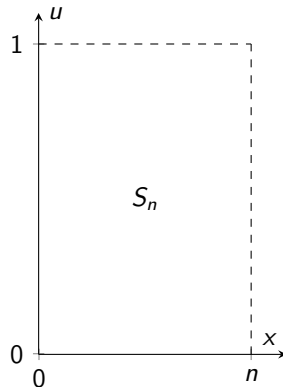
If $\gamma > 1/2$ and $\gamma' < 1/4$, then $\bar{S}_\infty(\cdot) \in D([0, 1], \mathbb{R})$, and

$$\bar{S}_n(\cdot) := n^{-\gamma} (S_n(t) - \mathbb{E}[S_n(t)]) \xrightarrow[n \uparrow \infty]{d} \bar{S}_\infty(\cdot) \quad \text{in } D([0, 1], \mathbb{R}).$$

Proof of Functional SLT

Step 1

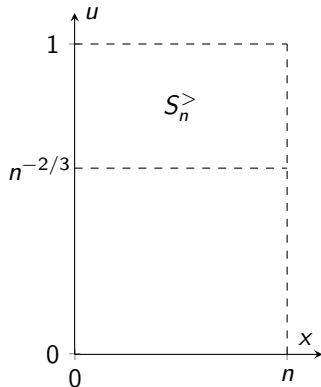
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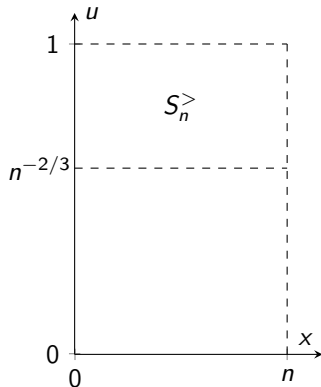
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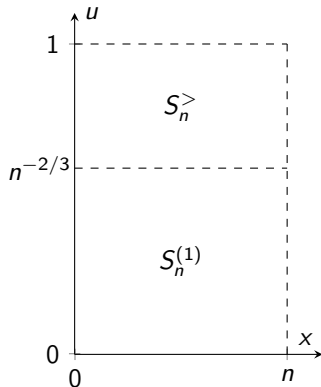
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Step 2

$$S_n^{(2)}(\cdot) := \sum_{P \in \mathcal{P} \cap \mathbb{S}_n} \mathbb{E}[\deg(P; \cdot) \mid P] \mathbb{1}\{U \leq n^{-2/3}\}$$

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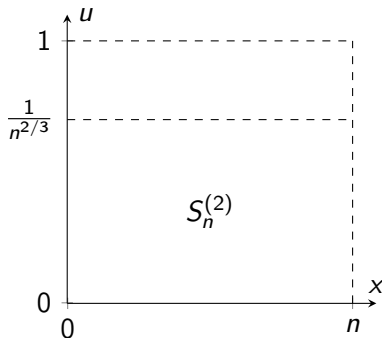
If $\gamma > 1/2$, then $\|\bar{S}_n^{(1)} - \bar{S}_n^{(2)}\| \xrightarrow[n \uparrow \infty]{\mathbb{P}} 0$.

Part 2 – Steps

$$\bar{s}_n^{(2)} \overset{\text{goal}}{\dashrightarrow} \bar{s}_\infty$$

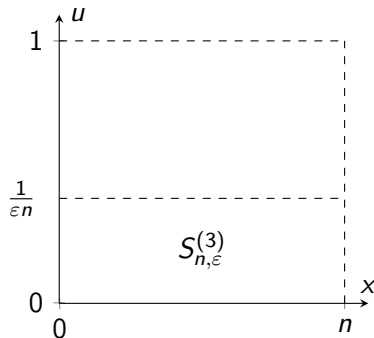
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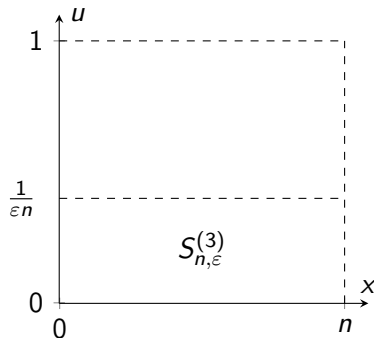
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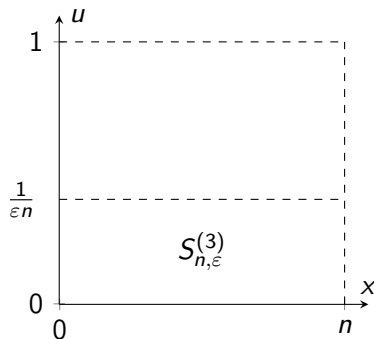
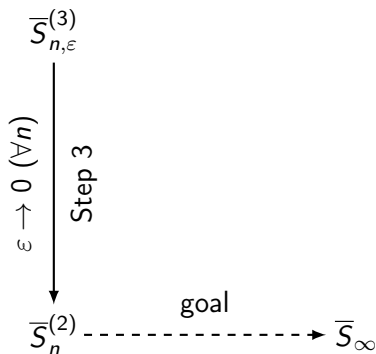
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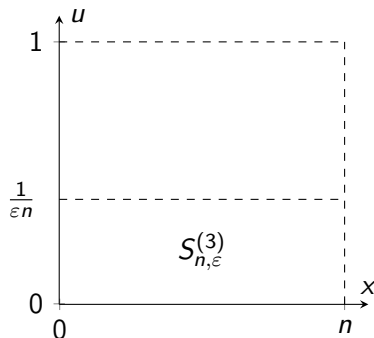
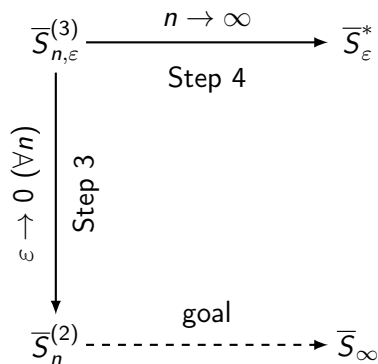
$$S_{n,\epsilon}^{(3)}(\cdot) := \sum_{P \in \mathcal{P} \cap \mathbb{S}_n} \mathbb{E}[\deg(P; \cdot) \mid P] \mathbb{1}\{U \leq 1/(\epsilon n)\}$$

Part 2 – Steps



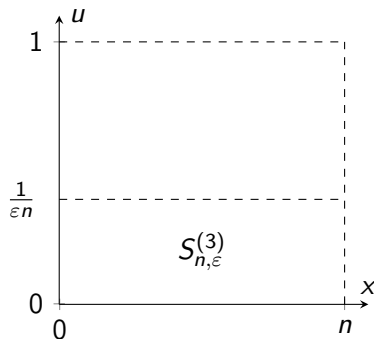
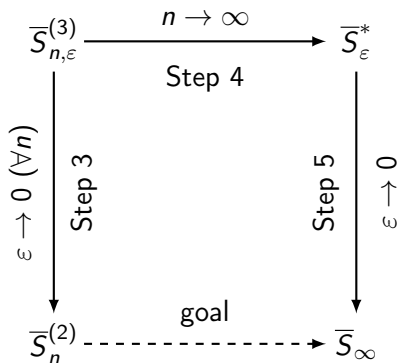
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Step 3

Goal:

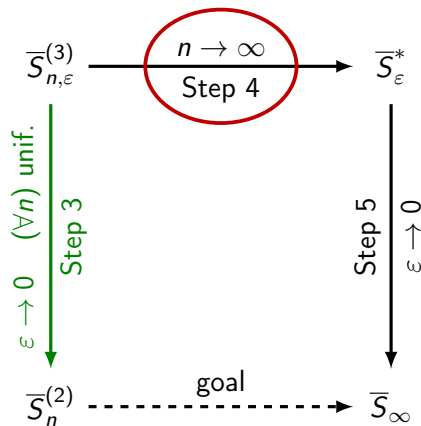
$$\lim_{\varepsilon \downarrow 0} \limsup_{n \uparrow \infty} \mathbb{P}(d_{\text{Sk}}(\bar{S}_{n,\varepsilon}^{(3)}, \bar{S}_n^{(2)}) > \delta) = 0 \quad \text{for all } \delta > 0$$

Proposition (Hirsch, Jahnel, J.; 2025+)

Let $\gamma > 1/2$ and $\gamma' \in (0, 1)$. Then,

$$\limsup_{n \uparrow \infty} \|\bar{S}_{n,\varepsilon}^{(3)} - \bar{S}_n^{(2)}\| \xrightarrow[\varepsilon \downarrow 0]{\mathbb{P}} 0.$$

Step 4



Step 4

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Lemma (Hirsch, Jahnel, J.; 2025+)

Let $\gamma > 1/2$ and $\gamma' \in (0, 1)$. Then,

$$n \mathbb{P}(n^{-\gamma} |N_s(P)| \in \cdot) \xrightarrow[n \uparrow \infty]{\nu} \nu(\cdot) \quad \nu([a, \infty)) := \tilde{c}^{1/\gamma} a^{-1/\gamma}.$$

S. I. Resnick. *Heavy-Tail Phenomena: Probabilistic and Statistical Modeling*. Springer, New York, 2007.

Step 4 – Convergence of Point Processes

$$\sum_{P \in \mathcal{P} \cap \mathbb{S}_n} \delta_{(n^{-\gamma}|N_{\bullet}(P_{\bullet})|, B, L)} \xrightarrow[n \rightarrow \infty]{\nu} \mathcal{P}_{\infty} = \text{PPP}(\nu \otimes \text{Leb}_{\mathbb{R}} \otimes \text{Exp}(1))$$

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 \downarrow \chi & & \\
 \bar{S}_{n,\varepsilon}^{(3)}(\cdot) = \sum_{P \in \mathcal{P} \cap \mathbb{S}_n} n^{-\gamma} |N_s(P_s)| (\cdot - B) & & \\
 \quad \times \mathbb{1}\{n^{-\gamma} |N_s(P_s)| \geq \tilde{c}\varepsilon^\gamma\} & & \\
 \quad \times \mathbb{1}\{B \leq \cdot \leq B + L\} & & \\
 - n^{-\gamma} \mathbb{E}[S_{n,\varepsilon}^{(3)}(\cdot)] & &
 \end{array}$$

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 \quad \times \mathbb{1}\{n^{-\gamma} |N_s(P_s)| \geq \tilde{c}\varepsilon^\gamma\} & & \quad \times \mathbb{1}\{J \geq \tilde{c}\varepsilon^\gamma\} \\
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Proposition (Hirsch, Jahnel, J.; 2025+)

χ is almost surely continuous with respect to the distribution of $\mathcal{P}_\infty$.

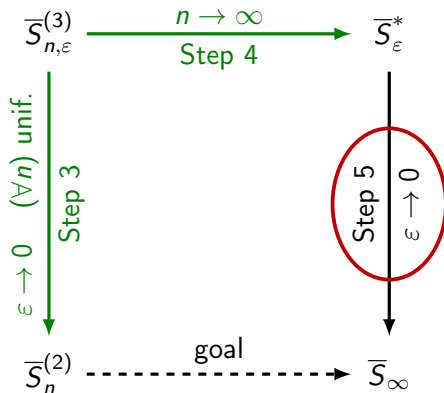
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Step 5



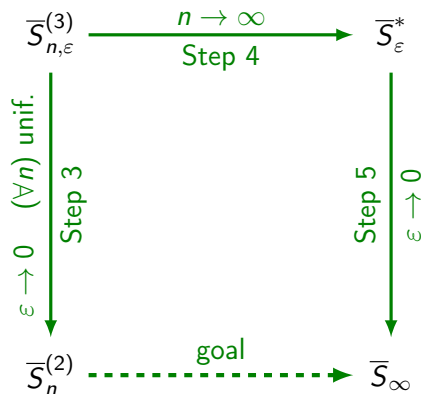
Step 5 – Almost Sure Convergence of $\overline{S}_\varepsilon^*$

Proposition (Hirsch, Jahnel, J.; 2025+)

Let $\varepsilon_k \rightarrow 0$ be a decreasing sequence. Then,

$$\sup_{n,m \geq N} \|\overline{S}_{\varepsilon_n}^* - \overline{S}_{\varepsilon_m}^*\| \xrightarrow[N \uparrow \infty]{\text{a.s.}} 0.$$

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Summary

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 - simple network model
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Thank you for your attention!

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References



M. Brun, C. Hirsch, P. Juhász, and M. Otto.
Random connection hypergraphs.
arXiv preprint arXiv:2407.16334v2, 2025.



C. Hirsch, B. Jahnel, S. K. Jhavar, and P. Juhász.
Poisson approximation of fixed-degree nodes in weighted random connection models.
Stochastic Process. Appl., 183:104593, 2025.



C. Hirsch, B. Jahnel, and P. Juhász.
Functional limit theorems for edge counts in dynamic random connection hypergraphs.
arXiv preprint arXiv:2507.16270, 2025.



C. Hirsch and P. Juhász.
Network simulator.
https://github.com/shepherd92/network_simulator, 2025.



C. Hirsch and P. Juhász.
On the topology of higher-order age-dependent random connection models.
Methodol. Comput. Appl. Probab., 27(2):Paper No. 44, 41, 2025.